

R. V. COLLEGE OF ENGINEERING® , Bengaluru-59
(Autonomous Institution under VTU)
VII Semester B. E. Examinations
Model Question Paper
MA375TGR – Mathematics of Music

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, one full question from 5 and 6, one full question from 7 and 8 & one full question from 9 and 10.

PART - A

1	1.1	A sound wave has an intensity of $I = 10^{-9} \text{ W/m}^2$. Find its sound level in decibels, show the calculation.	2
	1.2	State the differences between Shruti and Swara.	2
	1.3	What is the perfect fourth ratio in Just Intonation? Also, express it in cents.	2
	1.4	Write the continued fraction expansion of $\log_2(3/2)$ up to three terms.	2
	1.5	Define mātra, vibhāga, and āvartana in the context of Carnatic tāla system.	2
	1.6	What is the structure of Rupaka Tala? Express it using laghu, drutam, and anudruta notation.	2
	1.7	Give the Fourier series form of a periodic function and explain what the coefficients represent physically.	2
	1.8	Define the spectral centroid and compute it for a harmonic spectrum with amplitudes $A = \{5, 3, 1\}$ at frequencies $\{f, 2f, 3f\}$.	2
	1.9	What is self-similarity in fractal music? Give one example of algorithmic composition technique.	2
	1.10	What is a critical band in psychoacoustics? Why is it important for understanding consonance?	2

PART - B

2	a	Write down all the swaras in a) Carnatic system b) Hindustani system		5	
	b	If Sa (Shadja) = 240 Hz, calculate the frequencies of the following swaras using just intonation ratios: (5 marks)			5
		Swara	Ratio	Frequency	
		Sa	1:1	?	
		Ri	9:8	?	
		Ga	5:4	?	
Pa		3:2	?		
	Ni	15:8	?		
	c	Write a note on <i>Saptak</i> in Indian classical music.			6
3	a	Derive the equal temperament semitone ratio starting from first principles. Calculate the frequencies of the following notes in 12-tone equal temperament, given $A_4 = 440$ Hz: i) $A\#_4$ ii) C_5 iii) E_5 iv) A_5 v) B_5 vi) D_6			8
	b	A ‘C major 7 th ’ chord {C, E, G, B} is represented as {0, 4, 7, 11}. Transposing up a perfect fourth (5 semitones) gives: $T_5(\{0, 4, 7, 11\}) = \{5, _, _, _ \} = \{F, _, _, _ \}$ Also, invert it around axis 0.			8
		OR			
4	a	Prove that $\log_2(3/2)$ is irrational using proof by contradiction. Explain in detail the musical consequence of it.			8

	b	Derive the formula for (or explanation on) the total number of Melakarta rāgas (72) starting from the combinatorial possibilities of swaras, and elaborate on the <i>Katapayadi sutra</i> therein.	8
5	a	Explain the combinatorial approach to generating rhythmic variations. Discuss korvais and tihāis with mathematical patterns. Create one example of a 3-beat tihāi pattern.	8
	b	Explain the concept of Suladi Sapta Talas. Derive the 35 basic talas using the combination of 7 base talas and 5 jatis. Construct the tala for Tripura with Khanda jati and show its mathematical structure.	8
		OR	
6	a	Compare and analyze three different Carnatic tālas (Adi, Rupaka, Jhampa) in terms of their mathematical structure, beat patterns, and musical applications.	8
	b	Discuss how permutations and combinations are used in algorithmic composition of rhythmic patterns. Generate all possible 3-matra patterns using two types of durations (1 matra and 2 matra units). Show the combinatorial calculation.	8
7	a	For a square wave $f(t) = \begin{cases} k & \text{for } 0 < t < \frac{T}{2} \\ -k & \text{for } \frac{T}{2} < t < T \end{cases}$, derive its Fourier series representation showing that it contains only odd harmonics with amplitudes decreasing as $1/n$. Explain why the square wave sounds different from a sawtooth wave despite both being periodic.	8
	b	Derive the mathematical explanation for difference tones using the nonlinear distortion equation: $y = ax + bx^2$. If two pure tones at frequencies f_1 and f_2 are input, show that the output contains the difference frequency $ f_1 - f_2 $. When two notes at 600 Hz and 800 Hz are played simultaneously: (i) Calculate the first-order difference tone. (ii) Calculate the first-order summation tone. (iii) Identify the musical interval represented by 600 Hz and 800 Hz. Also, explain why combination tones are more prominent in organs and brass instruments.	8
		OR	
8	a	Prove that the set of all transposition operations $\{T_0, T_1, T_2, \dots, T_{11}\}$ forms a cyclic group under composition, isomorphic to $(\mathbb{Z}_{12}, +)$. Show that: (i) Closure: $T_m \circ T_n = T_{(m+n) \bmod 12}$ (ii) Associativity holds (iii) Identity element exists (T_0) (iv) Inverse elements exist	8
	b	Derive the Fourier series representation for a sawtooth wave. Explain the convergence of the series and how increasing the number of harmonics improves the approximation. Sketch the first three partial sums.	8
9	a	Discuss mathematical models of consonance and dissonance in music. Explain the role of frequency ratios and critical bands in determining consonant intervals.	8
	b	Define a first-order Markov chain. Construct a valid 3×3 transition matrix for generating melodies using notes {C, D, E} where stepwise motion has probability 0.5, staying on the same note has probability 0.3, and skipping has probability 0.2. Verify your matrix is row-stochastic.	8
10	a	Explain the missing fundamental phenomenon in psychoacoustics. A complex tone contains harmonics at 200 Hz, 300 Hz, 400 Hz, and 500 Hz but lacks the fundamental. What pitch will a listener perceive and why?	8
	b	Explain how Markov chains can be used for algorithmic music composition. Construct a simple first-order Markov model for melody generation with at least 4 pitch classes. Show the transition probability matrix.	8